ANALYTICAL CUMULATIVE EXAM

September 12, 1998

1. (25 points) The solute, A distributes between an organic and an aqueous phase according to the equilibrium

$$A_{(aq)} \Longleftrightarrow A_o$$

The equilibrium constant for the separation process, which is called the distribution coefficient, is given as

$$K_D = [A_o]/[A_{aq}]$$

Assuming that the original number of millimoles of A is x_o, **derive the following expression** which gives the number of millimoles of A, x_1 remaining in the aqueous phase after 1 extraction

$$x_1 = x_o(1/(1\ K_D V_o/V_{aq}))$$

In general, the fraction <u>remaining</u> in the aqueous phase after n extractions, x_n/x_o is given by

$$x_n/x_o = (1/(1\ K_D V_o/V_{aq}))^n$$

If x is in millimoles, then volumes are given in milliliters.

2. (25 points) Alternately, the distribution ratio, D is concerned with the extraction of all forms of a given species, for example, for the weak acid, HA

$$D = [HA]_o / ([HA]_{aq} + [A^-]_{aq})$$

Note that the relation above ignores A^- in the organic phase. Using the definition of D above, and the definition of K_a for the weak acid HA, **derive an expression** for D that contains only K_D, K_a, and $[H_3O^+]$.

3. (50 points) Two weak acids, HA and HB, have K_a values of 1×10^{-6} and 3×10^{-4}, respectively. Their distribution coefficients between water and ether are 12.2 for HA and 8.3 for HB. Design a separation procedure for the quantitative removal of HA (~99%) from a aqueous solution, while quantitatively retaining HB (~99%) in the aqueous phase. In situations where secondary chemical equilibria influence the separation, D can be substituted for K_D in the # expression above. In your experimental design you should specify (justify) an appropriate pH, and n. For simplicity, it is appropriate to set $V_{aq} = V_o$.